

Enabling Hybrid Poster Interaction via Face-to-Face Displays and Virtual Screen Sharing

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Abstract

Hybrid conferences have become more common since the COVID-19 pandemic, enabling onsite face-to-face interaction and remote participation via video. However, platforms like Zoom or Google Meet lack spatial and nonverbal cues, limiting remote users' sense of presence. We propose a hybrid poster presentation system using a 360-degree camera surrounded by 2×6 face-to-face displays to simulate mutual gaze and presence (Fig. 1). Remote HMD users view the onsite environment from a shared or movable viewpoint, with their faces shown on screens corresponding to their gaze direction, enabling natural online-online and online-onsite interactions. A virtual screen, placed in front of the real poster or projector screen, provides high-resolution content visibility for remote users. With 90% transparency, it also allows them to perceive the presenter's gestures, supporting more intuitive, gesture-aware communication.

CCS Concepts

• Human-centered computing;

Keywords

Face-to-face display, hybrid conference, gaze sharing, gesture sharing, telexistence, virtual reality.

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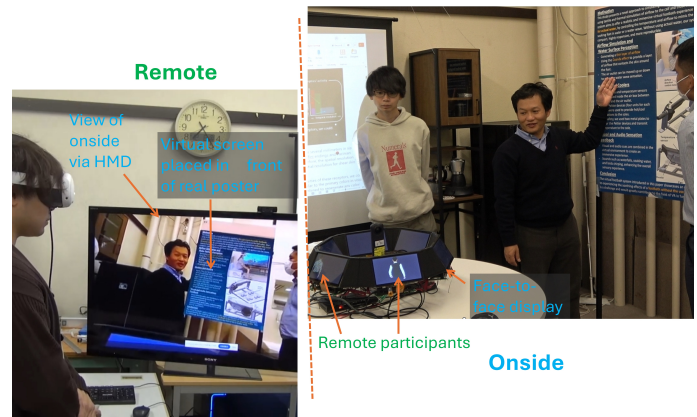


Figure 1: Hybrid poster presentation: Remote users wearing HMDs can freely view onsite posters, while presenters interactively explain their research to both side attendees.

1 Introduction

Hybrid academic conferences, particularly those involving poster presentations, have gained popularity since the COVID-19 pandemic to balance accessibility and interactivity. Poster sessions are an essential part of academic exchange, where multiple researchers gather around the presenter, engage in discussions, and visually explore detailed research results [Boggu 2015]. These sessions rely heavily on interactive, face-to-face communication, making hybrid adaptation particularly challenging.

Traditional platforms such as Zoom and Google Meet offer cost-effective access for remote participants but fail to recreate spatial cues like gaze direction or pointing gestures that are critical for communication [Monk and Gale 2002]. As a result, online participants lack presence in physical space, and onsite presenters cannot intuitively determine who is paying attention or which part of the poster is being observed. This makes it difficult to interact

with both onsite and remote audiences simultaneously, leading to reduced engagement and communication quality during hybrid poster sessions. Moreover, camera-based poster streaming often produces low-resolution views, making it hard for remote users to read fine text or identify regions being pointed to by the presenter. Onsite participants, in contrast, enjoy immediate visual and gestural feedback.

To address these limitations, our study introduces a face-to-face display system designed for hybrid poster presentations. The system links remote and onsite participants by rendering the gaze of remote users onto directional displays placed at poster stations. This setup allows onsite participants to immediately recognize who is looking at them or at specific parts of the poster. Remote users can share viewpoints or individually control their viewing direction using a mobile telepresence robot, enabling exploration interaction. Our approach allows onsite users to engage freely without wearing headsets, while preserving the spatial presence of remote users.

2 SYSTEM

2.1 Face-to-Face Display and 360-Degree Camera:

We used twelve Raspberry Pi 7-inch displays arranged in a circle around a 360-degree camera (Theta Z1, RICOH). Six of the displays faced inward toward the center (inward displays), while the other six faced outward (outward displays), as shown in Fig. 1. The inward displays are used to show the faces of remote users, while the outward displays can show their backs. Typically, remote users have only a single webcam, so an avatar can be used to represent their back view instead. The displays are controlled by a 2×6-channel video wall hardware system via a single HDMI input.

A 360-degree camera is used to live stream the onsite environment for remote users. When remote users join using a PC or HMD, their webcam feed and avatars are shown on the inward displays at positions corresponding to their viewing directions, allowing onsite participants to perceive who is looking at whom. This setup supports mutual face-to-face communication, where both remote and onsite users can recognize and respond to each other's gaze and facial expressions in a shared space.

Furthermore, the entire face-to-face display unit can be mounted on a telepresence robot, enabling remote users to change their viewpoint position within the onsite environment. This allows for a more immersive and self-directed interaction experience, simulating physical movement. While spherical displays have been explored for such purposes [Yonezawa et al. 2022], they often suffer from image distortion and do not easily support full face-to-face interaction between multiple remote users.

2.2 Real Poster and Virtual Screen:

In our system, a real printed poster is displayed onsite for in-person attendees, while a corresponding virtual screen is presented to remote participants. This approach addresses a major limitation of 360-degree cameras: their inability to capture the detailed text and visuals of posters or projector screens with sufficient clarity. Rather than relying solely on the 360 camera feed, a virtual screen is generated and displayed to remote users at the start of the presentation.

To support natural interaction, especially when presenters use gestures such as pointing to explain their work, the virtual screen is spatially aligned with the physical poster or projector screen. The virtual screen is rendered with approximately 90% transparency—a value chosen as a balance between clarity of the shared content and visibility of the presenter's gestures. At this transparency level, remote users can clearly view the virtual content while still being able to slightly perceive the motion of the presenter's arms behind the screen, as captured by the 360-degree camera. Although the presenter's full body may not be visible through the screen, this subtle motion helps remote users notice when and where gestures are occurring. Without this motion visibility, gestures can easily go unnoticed.

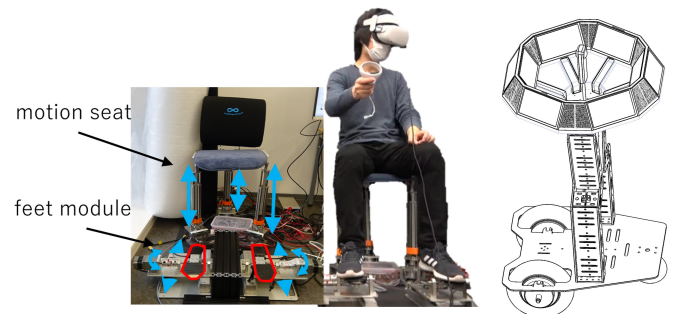


Figure 2: An attendee sits on a motion chair and controls a telepresence robot to change the viewpoint position.

3 DEMONSTRATION

An attendee will participate as a remote HMD user, operating a telepresence robot to explore a hybrid poster session (Fig. 2). A presenter onsite will stand by a physical poster and interact via the face-to-face display. Remote user sees a high-resolution virtual poster screen aligned with the real one, with the presenter's gestures visible through a semi-transparent overlay. Our motion feedback platform provides waist and leg sensations, enhancing immersion. This demo builds on our prior work [Kikuchi et al. 2022], by applying immersive motion and viewpoint sharing to hybrid presentations.

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